ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO5 – CASE STUDY BASED ASSIGNMENT

DWDM Metro Link Commissioning: From Field Measurements to a Standards-Based Upgrade Decision

Student Name	P. ROHITH
Register Number	192312015
Course Code & Title	ECA18 – Optoelectronics and Optical Communication
Course Outcome	CO5
Nature of Assessment	Case Study Based Assignment
Duration	1 Week
Maximum Marks	100
Primary Bloom's Levels	K4 – Analyze | K5 – Evaluate | K6 – Create/Design
Knowledge Profile	WK1, WK3, WK5

Case Study Title

DWDM Metro Link Commissioning: From Field Measurements to a Standards-Based Upgrade Decision

Engineering Objective

Develop an integrated diagnostic, repair and upgrade configuration for a 60 km G.652.D DWDM metro link using field power measurements, OSA/OSNR evidence, OTDR traces, eye-diagram and BER diagnostics, quantum-limit analysis and ITU-T G.694.1 channel planning.

Overall Decision Requirement

The repaired link must meet the present 10 Gb/s communication requirement with at least 3 dB receiver margin and BER ≤ 10^-12, while providing a defensible path to 1.6 Tb/s on the same fibre pair without re-laying the fibre.

Report Structure

REQUIREMENT → MEASUREMENT → ANALYSIS → ROOT-CAUSE IDENTIFICATION → REPAIR → UPGRADE → VERIFICATION → DEPLOYMENT DECISION

EXECUTIVE SUMMARY & DESIGN REQUIREMENT STATEMENT

1. Executive Summary

A metro operator has commissioned a 60 km ITU-T G.652.D link between two data centres. Eight 10 Gb/s NRZ channels are carried on a 100 GHz ITU-T G.694.1 grid, with +1 dBm launch power per channel. The line uses a transmit booster EDFA, and acceptance testing has revealed two different classes of problems: Channel 3 has a healthy eye and Q = 7.6, while Channel 5 has a partially closed eye, Q = 4.7 and ER = 4.5 dB. In parallel, the field loss budget is worse than the design allowance.

The supplied field data identify Event 2 at 31.50 km as a 0.62 dB mechanical splice and Event 3 at 59.90 km as a 1.20 dB reflective LC/UPC patch-panel connector. Event 4 occurs only 9 m after Event 3 and is another reflective connector. The measured loss is therefore not just a theoretical budget issue; it contains identifiable repairable defects.

The engineering analysis shows that the as-built passive loss is 22.52 dB, compared with a design allowance of 21.25 dB. At +1 dBm launch power, the raw received power is -21.52 dBm, giving only 2.48 dB above the -24 dBm receiver sensitivity. This is below the required 3 dB margin even before channel-dependent penalties.

The OSNR calculation specified by the case gives 47 dB at the receiver, well above the 20 dB requirement. PMD is 0.775 ps and is therefore not the limiting impairment at 10 Gb/s. The most important actions are consequently physical-loss reduction and Channel 5 transmitter correction.

2. Design Requirement Statement

The repaired and upgraded 60 km link shall: (1) use the existing G.652.D fibre pair; (2) provide received power at least 3 dB above -24 dBm sensitivity; (3) achieve BER ≤ 10^-12 and Q ≥ 7.03; (4) maintain ER ≥ 8.2 dB for the specified receiver sensitivity validity; (5) maintain OSNR ≥ 20 dB in 0.1 nm; (6) control dispersion and PMD; (7) remove excessive OTDR events; and (8) support 1.6 Tb/s using a standards-based WDM configuration with improved spectrum and energy efficiency.

STAGE 1 – INTERPRET THE MEASUREMENT / PERFORMANCE REQUIREMENT

3.1 Measurement Instruments and Their Roles

Instrument	Quantity validated	Commissioning purpose
Optical Power Meter	Launch/received power; insertion loss	Confirms actual optical budget and receiver margin.
OSA	Channel wavelength, channel power, OSNR	Checks WDM grid, power flatness and ASE/noise performance.
OTDR	Event position, loss, reflectance	Localizes splice/connector defects and verifies repair.
Eye / BER Tester	Eye opening, Q, BER, ER, DJ, RJ	Determines digital signal quality and target compliance.
Dispersion / PMD Analyzer	Chromatic dispersion and PMD	Checks pulse broadening and polarization impairment.

3.2 Loss, Penalty and Noise Components

- Fibre attenuation: $\alpha = 0.21$ dB/km at 1550 nm over 60 km.
- Demultiplexer insertion loss: 6.0 dB.
- Connector losses: measured field values replace the nominal design allowances.
- Splice losses: measured field values replace the nominal 0.05 dB assumption where applicable.
- Chromatic-dispersion penalty: evaluated from the supplied δ relation.
- Extinction-ratio penalty: especially important for Channel 5 because ER = 4.5 dB is below the specified 8.2 dB condition.
- EDFA ASE noise: evaluated through the supplied OSNR expression.

3.3 Eye, Q, BER and Jitter Interpretation

A clearly open eye indicates adequate vertical and horizontal decision margin. A partially closed eye can result from noise, low extinction ratio, deterministic jitter and random jitter. The Q-factor is directly linked to BER in the supplied relation. The target Q is 7.03 for BER = 10^-12. Therefore Channel 3 passes with Q = 7.6, while Channel 5 fails with Q = 4.7.

3.4 OTDR Physics

OTDR dynamic range limits the maximum observable link loss. Event dead zone determines whether adjacent reflective events can be separated, while attenuation dead zone determines whether the loss following a strong event can be measured. Since Events 3 and 4 are separated by only 9 m, a short pulse is needed for local diagnosis.

3.5 Constraints

The engineering solution is bounded by the existing fibre pair, the ≥3 dB margin requirement, the 1.6 Tb/s target, G.694.1 spectrum planning, OSNR requirements and energy efficiency. Fibre re-laying is not permitted.

STAGE 2 – LINK BUDGET, DISPERSION AND OSNR

4.1 Fibre Attenuation

Fibre loss = αL = 0.21 × 60 = 12.60 dB.

4.2 Design Versus As-Built Loss

Component	Design allowance	Measured / as-built
60 km fibre	12.60 dB	12.60 dB
Demultiplexer	6.00 dB	6.00 dB
4 connectors	4 × 0.5 = 2.00 dB	2.70 dB total measured connector contribution
13 splices	13 × 0.05 = 0.65 dB	1.22 dB total measured splice contribution
TOTAL	21.25 dB	22.52 dB

4.3 As-Built Received Power

P_rx = P_tx – L_total = +1.0 – 22.52 = -21.52 dBm.

The receiver sensitivity is -24 dBm and the required design margin is 3 dB. Therefore the minimum acceptable received power is -21 dBm. The raw as-built value of -21.52 dBm is 0.52 dB below this requirement.

Raw power margin = -21.52 - (-24) = 2.48 dB. Required margin = 3 dB. Margin shortfall = 0.52 dB before additional penalties.

4.4 Excess Loss Identification

The total measured loss exceeds the design value by 1.27 dB. The most significant discrete contributions are the 1.20 dB Event 3 connector and the 0.62 dB Event 2 mechanical splice. These are the first physical defects to correct because they can be directly removed or reduced by repair.

4.5 OSNR

Using the supplied relation OSNR = 58 + P_in – NF, with P_in = -6 dBm and NF = 5 dB: OSNR = 58 - 6 - 5 = 47 dB. This exceeds the required 20 dB by 27 dB. Hence the supplied OSNR model does not indicate a noise-limited link.

4.6 Engineering Interpretation

The link is primarily constrained by power margin and discrete physical loss rather than by OSNR or PMD. Increasing amplifier gain before repairing the fibre would not be the preferred first action because it does not remove the measured connector/splice defects and can increase ASE.

STAGE 2 – DISPERSION, PMD AND LINK-BUDGET DESIGN DECISION

5.1 Chromatic Dispersion

The case gives D = 17 ps/(nm·km), L = 60 km and source rms spectral width σλ = 0.03 nm. The product BDLσλ for B = 10 Gb/s is 10^10 × 17 × 60 × 0.03 ps after consistent unit conversion, corresponding to 306 ps of first-order broadening.

The supplied dispersion-penalty expression is δ = 5 log10[1 + 2π(BDLσλ)^2]. Substitution indicates that dispersion cannot be ignored in the analytical budget. In a practical deployment, the transmitter spectral width, actual dispersion slope and receiver tolerance must be verified rather than assuming that a raw power margin alone guarantees BER.

5.2 PMD

PMD = D_PMD√L = 0.1√60 = 0.775 ps.

At 10 Gb/s, T_b = 1/B = 100 ps. The 10% tolerance is 10 ps. Since 0.775 ps < 10 ps, PMD is safely within the stated 10 Gb/s tolerance.

5.3 Channel-Specific Extinction-Ratio Penalty

Parameter	Channel 3	Channel 5
Q	7.6	4.7
ER	9.0 dB	4.5 dB
DJ	0.18 UI pk-pk	0.30 UI pk-pk
RJ	0.012 UI rms	0.022 UI rms
Eye condition	Clearly open	Partially closed
BER target compliance	Pass	Fail

Channel 3 meets ER ≥ 8.2 dB. Channel 5 is far below this threshold. Because both channels share the same 60 km fibre, this difference is strong evidence of a channel-specific transmitter or channel path impairment.

5.4 Design Decision 1

I recommend repairing the physical excess loss first, because the as-built loss is 22.52 dB and the raw margin is only 2.48 dB. The dominant defects are Event 3 and Event 2. After repair, remeasure received power and OSNR. Launch power should only be adjusted after physical loss is minimized and the actual post-repair margin is known.

STAGE 3 – OTDR, SPLICE AND CONNECTOR CONFIGURATION

6.1 OTDR Dynamic-Range Requirement

The case specifies the required dynamic range as one-way trace loss + 6 dB. Using 22.52 dB as the as-built loss gives approximately 28.52 dB required dynamic range.

Pulse width	Dynamic range	Event dead zone	Attenuation dead zone	Use
10 ns	20 dB	1 m	4 m	Local high-resolution event diagnosis
100 ns	27 dB	5 m	15 m	Intermediate reach; not ideal for 9 m separation
1 µs	33 dB	40 m	100 m	Full-link reach; poor event resolution

The 1 µs pulse is the only supplied setting that exceeds the approximately 28.52 dB dynamic-range requirement. However, its dead zones are too large to separate the two connectors 9 m apart. Therefore a two-setting OTDR procedure is required: 1 µs for end-to-end reach and 10 ns for local event characterization.

6.2 Spatial Pulse Length

Propagation velocity v = c/n = 2.998×10^8 / 1.4682 ≈ 2.042×10^8 m/s.

Pulse	One-way spatial pulse length
10 ns	≈ 2.04 m
100 ns	≈ 20.42 m
1 µs	≈ 204.2 m

The 9 m separation between Events 3 and 4 is larger than the 10 ns spatial pulse length but smaller than the 100 ns and 1 µs values. Thus 10 ns is the appropriate pulse for attempting to distinguish the two far-end connectors.

6.3 OTDR Test Arrangement

- Test from both ends whenever possible to avoid masking or ghost-event interpretation.
- Use launch and receive fibres so the first and last connectors can be characterized correctly.
- Use a 1 µs pulse for full-link reach and 10 ns for close-event localization.
- Repeat the trace after each repair and compare event loss/reflectance with the pre-repair baseline.

STAGE 3 – EVENT ANALYSIS AND REPAIR PRIORITY

7.1 OTDR Event Interpretation

Event	Location	Loss	Reflectance	Engineering interpretation
1	12.00 km	0.05 dB	Non-reflective	Normal fusion splice; no corrective action.
2	31.50 km	0.62 dB	-34 dB	Excessive mechanical splice loss; replace with fusion splice.
3	59.90 km	1.20 dB	-22 dB	Major reflective/high-loss LC/UPC patch-panel connector; highest priority.
4	59.909 km	Not resolved	-40 dB	Second patch-cord connector 9 m after Event 3; verify after Event 3 repair.

7.2 Repair Priority

Priority 1: Event 3. Its 1.20 dB loss is the largest single measured event and its -22 dB reflectance indicates a significant reflective discontinuity. Priority 2: Event 2. Its 0.62 dB loss is much larger than the 0.05 dB fusion-splice performance observed elsewhere. Event 4 should be inspected after Event 3 because its loss is unresolved due to the close spacing and dead-zone limitations.

7.3 Fusion Splice Selection

A properly prepared fusion splice is preferred for Event 2 because it produces a permanent joint with low insertion loss and low reflection. The twelve fusion splices at 0.05 dB provide the field benchmark. The mechanical splice at 0.62 dB is therefore clearly anomalous.

7.4 Connector Selection

The existing far-end event is an LC/UPC patch-panel connector. A replacement LC interface matching the panel is the most practical choice. UPC is appropriate where the existing equipment requires UPC mating. APC can be selected where lower back-reflection is required, but only with compatible APC hardware. MPO and CS are high-density alternatives but are not appropriate as a blind replacement for an existing LC/UPC panel.

7.5 Round-Trip Delay

Using $t = 2nL/c$: Event 1 at 12 km ≈ 117.55 µs; Event 2 at 31.5 km ≈ 308.57 µs; Event 3 at 59.90 km ≈ 586.72 µs; Event 4 at 59.909 km ≈ 586.81 µs.

7.6 Location Uncertainty

For ±0.002 uncertainty in group index, fractional distance uncertainty is ±0.002/1.4682 ≈ ±0.1362%. This gives approximately ±16.3 m at 12 km, ±42.9 m at 31.5 km and ±81.6 m at 59.9 km. Absolute distance should therefore be combined with event-pattern evidence.

STAGE 4 – EYE DIAGRAM, BER AND QUANTUM-LIMIT ANALYSIS

8.1 BER from Q

BER = 0.5 erfc(Q/√2). For Q = 7.6, BER ≈ 1.48×10^-14. For Q = 4.7, BER ≈ 1.30×10^-6.

Channel	Q	BER	Required BER	Result
3	7.6	1.48×10^-14	≤10^-12	PASS
5	4.7	1.30×10^-6	≤10^-12	FAIL

8.2 Total Jitter

At BER = 10^-12, TJ = DJ + 14.07RJ.

Channel	DJ	RJ	TJ	Approx. eye width = 1-TJ
3	0.18 UI	0.012 UI	0.34884 UI	0.65116 UI
5	0.30 UI	0.022 UI	0.60954 UI	0.39046 UI

Channel 5 has almost twice the total jitter of Channel 3 and consequently much smaller horizontal eye opening.

8.3 Vertical Versus Horizontal Margin

The low ER of Channel 5 reduces vertical opening because the logical '1' and '0' levels are less separated. The larger DJ and RJ reduce horizontal opening. Therefore both amplitude and timing margins are impaired.

8.4 Root Cause

The common fibre cannot by itself explain the large channel-to-channel difference because Channel 3 is healthy on the same route. Channel 5's ER = 4.5 dB is below the receiver's stated ER-validity condition of 8.2 dB. The primary corrective action is therefore Channel 5 transmitter inspection, bias/modulation correction or transmitter replacement.

8.5 Acceptance Evidence

- Q ≥ 7.03.
- BER ≤ 10^-12.
- ER ≥ 8.2 dB.
- Eye visibly open with adequate vertical and horizontal margin.
- TJ reduced to a value consistent with the receiver's timing budget.

STAGE 4 – QUANTUM LIMIT AND DETECTION COMPARISON

9.1 Ideal Direct-Detection Limit

The supplied direct-detection relation is BER = 0.5 exp(-N). For BER = 10^-12:

N = ln(0.5/10^-12) = ln(5×10^11) ≈ 26.94 photons/bit.

At λ = 1550 nm, photon energy E = hc/λ = (6.626×10^-34)(2.998×10^8)/(1.55×10^-6) ≈ 1.282×10^-19 J.

At 10 Gb/s, the ideal quantum-limited power is P = NEB ≈ 26.94 × 1.282×10^-19 × 10^10 = 3.45×10^-8 W ≈ -44.62 dBm.

9.2 Practical Receiver Gap

The specified receiver sensitivity is -24 dBm. The ideal direct-detection quantum limit is approximately -44.62 dBm. Therefore the practical receiver is approximately 20.62 dB above the ideal photon-counting limit.

This gap is expected because practical systems include detector noise, extinction-ratio limitations, timing jitter, implementation penalties, finite bandwidth and other non-ideal effects.

9.3 Homodyne and Heterodyne PSK

For homodyne PSK, BER = 0.5 erfc(√(2N)). For heterodyne PSK, BER = 0.5 erfc(√N). Solving for BER = 10^-12 gives approximately N = 12.37 photons/bit for homodyne and N = 24.74 photons/bit for heterodyne.

Detection method	Photons/bit at BER 10^-12	Approx. ideal sensitivity at 10 Gb/s
Direct detection	26.94	-44.62 dBm
Homodyne PSK	12.37	≈ -47.99 dBm
Heterodyne PSK	24.74	≈ -44.99 dBm

9.4 Implication for Coherent WDM

Coherent detection provides phase-sensitive detection and supports higher-order modulation formats such as DP-QPSK and DP-16QAM. However, practical coherent receivers require adequate OSNR and incur implementation penalties. Therefore the quantum limit is a theoretical reference, not the sole commissioning criterion.

Design Decision 3

Channel 5 should not be accepted until ER, Q, BER and eye opening are restored. The common fibre should not be replaced merely because Channel 5 fails; the measurement evidence points to a channel-specific impairment.

STAGE 5 – G.694.1 WDM UPGRADE AND SPECTRAL EFFICIENCY

10.1 50 GHz Grid Calculation

Using the 193.1 THz anchor and a 50 GHz grid, channel n = +3 has f = 193.1 + 3(0.05) = 193.25 THz.

Using λ = c/f, λ ≈ 2.998×10^8 / 193.25×10^12 = 1.55136×10^-6 m ≈ 1551.36 nm.

Parameter	Calculated value
Grid anchor	193.1 THz
Grid spacing	50 GHz
Channel index	+3
Channel frequency	193.25 THz
Approximate wavelength	1551.36 nm

10.2 Option P – 16 × 100G DP-QPSK

Capacity = 16 × 100G = 1.6 Tb/s. Fixed spectrum = 16 × 50 GHz = 800 GHz. Payload spectral efficiency = 1.6/0.8 = 2.0 bit/s/Hz. Required OSNR = 12.5 dB. Module power = 16 × 22 W = 352 W.

10.3 Option Q – 8 × 200G DP-16QAM

Capacity = 8 × 200G = 1.6 Tb/s. With 32 GBd and roll-off 0.1, occupied bandwidth per channel is approximately 32 × 1.1 = 35.2 GHz. At 12.5 GHz slot granularity, ceil(35.2/12.5) = 3 slots per channel. Eight channels require 24 slots = 300 GHz. Payload spectral efficiency on the allocated slots is 1.6/0.3 ≈ 5.33 bit/s/Hz. Required OSNR = 19.5 dB. Module power = 160 W.

10.4 Option R – 4 × 400G Si-Photonic Pluggables

Capacity = 4 × 400G = 1.6 Tb/s. Fixed spectrum = 4 × 75 GHz = 300 GHz. Payload spectral efficiency ≈ 5.33 bit/s/Hz. Required OSNR = 26 dB. Module power = 4 × 15 = 60 W.

STAGE 5 – ENERGY COMPARISON AND INTEGRATED DESIGN

11.1 Energy per Bit

Option	Total capacity	Total module power	Energy/bit
P	1.6 Tb/s	352 W	0.220 pJ/bit
Q	1.6 Tb/s	160 W	0.100 pJ/bit
R	1.6 Tb/s	60 W	0.0375 pJ/bit
Legacy 10G SFP+	10 Gb/s	3.5 W	0.350 pJ/bit

11.2 Integrated Comparison

Criterion	P	Q	R
Spectrum	800 GHz	300 GHz	300 GHz
Spectral efficiency	2.0	5.33	5.33
OSNR requirement	12.5 dB	19.5 dB	26 dB
Module power	352 W	160 W	60 W
Energy/bit	0.220 pJ/bit	0.100 pJ/bit	0.0375 pJ/bit
Engineering assessment	Power/spectrum inefficient	Best balance	Best energy, highest OSNR demand

11.3 Selection of Option Q

Option R has the lowest module energy, but its 26 dB OSNR requirement is substantially higher than Option Q. Option P is easier from an OSNR perspective but uses 800 GHz and 352 W. Option Q provides a strong balance: 1.6 Tb/s, approximately 300 GHz allocated spectrum, 5.33 bit/s/Hz, 19.5 dB OSNR requirement and 160 W total module power.

The selection is therefore based on the integrated objective rather than a single metric. Q is the preferred deployment choice because it combines high spectral efficiency with lower energy than P and a less demanding OSNR target than R.

11.4 Si-Photonic Integration

The Si-photonic option illustrates the energy advantage possible through integration and reduced module count. Fewer high-capacity modules can lower power, rack space and interconnect complexity. The trade-off is that the optical performance and OSNR budget must remain adequate for the selected modulation format.

11.5 Sequence of Work

- 1. Repair Event 3 and Event 2.
- 2. Re-run OTDR and optical power measurements.
- 3. Correct Channel 5 transmitter and repeat eye/BER measurements.
- 4. Confirm OSNR across all channels.
- 5. Plan the 12.5 GHz flex-grid allocation for Option Q.
- 6. Commission 8 × 200G coherent channels.
- 7. Repeat OSA, power, OSNR, BER and eye measurements.

STAGE 6 – FINAL VERIFICATION, DESIGN AND DEPLOYMENT DECISION

12.1 Final Verification Table

Requirement	Target	Post-repair acceptance criterion	Status
Total optical loss	≤ 21.25 dB design	Reduce measured 22.52 dB through repairs	Conditional
Received power	≥ -21 dBm	≥ -21 dBm per channel	Verify
System margin	≥ 3 dB	≥ 3 dB	Verify
OTDR events	No excessive event	Event 2/3 loss reduced; Event 4 characterized	Verify
Eye / Q / ER	Q ≥ 7.03; ER ≥ 8.2 dB	All channels satisfy	Verify
BER	≤ 10^-12	All channels satisfy	Verify
PMD	≤ 10% bit period	0.775 ps < 10 ps	PASS
OSNR	≥ 20 dB	47 dB by supplied model	PASS
Capacity	1.6 Tb/s	8 × 200G = 1.6 Tb/s	PASS after commissioning
Energy	Minimize pJ/bit	Option Q = 0.100 pJ/bit	Preferred

12.2 Final Diagnostic / Upgrade Workflow

REQUIREMENT → POWER/OSA MEASUREMENT → AS-BUILT BUDGET → OTDR 1 µs REACH TEST → OTDR 10 ns LOCAL TEST → EVENT 3 CONNECTOR REPAIR → EVENT 2 FUSION SPLICE → OTDR + POWER VERIFICATION → CHANNEL 5 TRANSMITTER CORRECTION → EYE/Q/BER/ER VERIFICATION → G.694.1 FLEX-GRID PLANNING → OPTION Q COHERENT COMMISSIONING → FINAL OSA/POWER/BER TEST → DEPLOYMENT.

12.3 Final Engineering Decision

YES – WITH TECHNICAL CONDITIONS. The existing fibre pair can be used for the intended 1.6 Tb/s upgrade without re-laying the fibre, but deployment must be conditional on successful physical repair and Channel 5 restoration.

The largest physical defect is Event 3 at 59.90 km with 1.20 dB loss and -22 dB reflectance. Event 2 at 31.50 km contributes 0.62 dB and is a mechanical splice. Both are repairable. The present as-built loss is 22.52 dB and gives -21.52 dBm received power at +1 dBm launch, corresponding to only 2.48 dB raw margin. Therefore repair is mandatory before final acceptance.

Channel 3 meets the BER target, with BER ≈ 1.48×10^-14. Channel 5 fails badly, with BER ≈ 1.30×10^-6. Its ER = 4.5 dB is below the 8.2 dB condition and its total jitter is approximately 0.610 UI. The transmitter must be corrected or replaced.

The supplied OSNR model gives 47 dB, while PMD is 0.775 ps. These results indicate that neither OSNR nor PMD is the immediate limiting factor. The engineering priority is therefore physical-loss repair followed by channel-specific signal correction.

Option Q is selected for the upgrade because it provides 1.6 Tb/s using 8 × 200G channels, approximately 300 GHz of flex-grid allocation, about 5.33 bit/s/Hz payload spectral efficiency and 0.100 pJ/bit module energy. It is a balanced choice between spectrum, OSNR requirement and power.

12.4 Conclusion

The complete case-study process demonstrates that reliable DWDM commissioning requires correlation of multiple measurements rather than reliance on one instrument. OTDR identifies where the physical losses occur, the power meter establishes the link budget, the OSA verifies channel and OSNR behavior, and the eye/BER tester identifies the Channel 5 impairment. Quantum-limit calculations provide a theoretical sensitivity reference, while G.694.1 planning enables a standards-based capacity upgrade.

Final recommendation: repair Event 3 first, replace Event 2 with a fusion splice, characterize Event 4 with the appropriate short-pulse OTDR method, correct Channel 5 until Q ≥ 7.03 and ER ≥ 8.2 dB, verify ≥3 dB system margin and OSNR ≥20 dB, and then deploy Option Q for the 1.6 Tb/s upgrade.